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Towards a semantic structure for classifying IoT agriculture sensor datasets : An approach based on machine learning and web semantic technologies

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ABSTRACT

With the increase in the number of IoT farming datasets, it has become so difficult to identify the right data for IoT agriculture applications. Therefore, a meaningful structure is needed to well understand, interpret and index IoT farming datasets. This paper proposes a new IoT farming ontology that allows the organization, the understanding, and the classification of IoT agriculture datasets knowledge as well as meta-data storage. For this, we have developed a new IoT agriculture taxonomy that helps to identify an IoT agriculture application based on the combination of various IoT agriculture sensors. The evaluation of the semantic IoT agriculture datasets classification, based on the background knowledge provided by the proposed ontology, was achieved using Machine Learning algorithms, including Logistic Regression, Decision Tree Classifier, K-Neighbors Classifier, Linear Discriminant Analysis, Gaussian NB, SVM, and Random Forest Regressor. The obtained results clearly show the effectiveness of the proposed ontology to classify IoT agriculture datasets with high performances and accuracy (0.98), (0.99) using Decision tree classifier and SVM respectively.

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1. Introduction

In the last decade, the IoT has become an essential tool in several domains. In agriculture, IoT aims to develop connected agricultural systems that can maximize yields using minimal resources. Agriculture implements IoT through the use of IoT sensors. An IoT sensor is a smart device that can sense anomalies and transfer data over a network. A wide range of IoT sensors are used in agriculture (i) installed throughout the field (Jamroen et al., 2020; Phasinam et al., 2022 and Deshpande and Lokare, 2019), including soil moisture sensors, soil temperature sensors, pH sensors, etc. (ii) in the IoT systems (iii) on smart agriculture machines (iv) on weather station (Pelosi et al., 2020) including air humidity sensors,

air temperature sensors, precipitation sensors and dew points sensors, etc. Data collected from the IoT agriculture sensors can be used to monitor crop growth, to build models, to predict crop behaviour under some conditions, to reduce water consumption and to automate the irrigation system. It can also be used to monitor and to control the climate in greenhouses. The use of IoT in greenhouse farming can reduce the need of human intervention and manual monitoring.

Speaking about data collected from IoT agriculture sensors indirectly involves IoT agriculture datasets. IoT agricultural datasets can take many forms: structured, semi-structured, and unstructured form (Cravero et al., 2022 and Zeginis et al., 2019). For many years, researchers have re-used publicly available repositories in their applications (Rosli et al., 2016).

The users of the applications need to extract the exact knowledge from a proper dataset by exploring, analyzing, and classifying various datasets. The classification of datasets by domain has already been studied (Gregory et al., 2020; Wang et al., 2020 and Wang et al., 2022); and has been proven to be very important since the topical relevance is an interesting factor to decide for secondary data (Rosli et al., 2016). This paper proposes a classification of datasets in the field of agriculture. It focuses on how to select the right IoT sensor dataset which suits the needed application from an

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existing IoT agricultural datasets even without available guiding information (Fig. 1).

This issue is neither how to gather, nor how to treat the stored data, it is rather how to represent and classify the IoT agricultural datasets in order to facilitate their re-use.

Selecting an appropriate and high-quality dataset is one of the most challenging tasks. The availability of a large number of IoT agriculture datasets with easy access to the public makes this task even harder. Despite that the production of datasets is with a high volume, fast velocity, and diverse variety of data collected from the environment, the IoT sensors datasets often lack pertinent information which can help users to select the appropriate and high-quality dataset. In addition, dataset analysis leads users to extract properties, characteristics, or elements with no or a weak semantic value in the majority of cases which is another significant hurdle. Indeed, preserving both the data quality and semantics is a challenging task.

This paper aims to address these issues based on ontologies which can give a semantic structure to represent elements in IoT agriculture datasets. This process allows machine learning algorithms to easily classify datasets by IoT agriculture applications. Ontology is a powerful tool that combines semantics with the standardized description formats to merge and exchange heterogeneous information (Shi et al., 2018). In our context, the Ontology Web Language (OWL) is used to model agriculture domain knowledge and IoT agriculture sensor datasets elements to make datasets easier to handle for various IoT agriculture applications.

Recently, ontologies are becoming more and more popular in the Machine Learning (ML) domain as a source of background knowledge (Kulmanov et al., 2020). This knowledge can be used in ML algorithms for two purposes: (i) to enhance the set of features to be exploited, (ii) to restrict the search for the best solution to a learning problem (Kulmanov et al., 2020).

ML algorithms are used for a large range of tasks including classification, regression, clustering, predictions, pattern recognition, and association rules determination (Zantalis, 2019) and exploited in a variety of IoT applications such as healthcare analytics, environmental pollution management, economics and precision agriculture (Rogers and Girolami, 2016; Malik et al., 2018; Ha et al., 2020; Akhter and Sofi, 2022).

Applying machine learning techniques has become a trend to overcome the reasoning hardness for the pair ontology/reasoner. In (Kang et al., 2014), authors developed predictive models to predict performance on ontology classification using regression techniques, with ontology metrics as features. The results showed that, the predicted models achieve high accuracy, good generalization with a statistical significance. In (Pan et al., 2018), the Random Forest algorithm was proposed to predict and analyse ontology reasoning in OWL 2 EL ontologies based on the notion of ABox Intensity. The results showed that this notion has a strong influence on the predictability of the performance of ontology reasoning. In (Liu et al., 2019), authors associate sensor data with existing domain ontologies. They proposed a mechanism that uses a random forest-based learning model to classify the sensor instances in sensor domain ontologies. In both studies Random Forest based regression models showed a consistent high accuracy.

The main objective of this paper is to propose a system that allows IoT agriculture applications to classify IoT agriculture sensors datasets. Firstly, a taxonomy was proposed to identify IoT agriculture sensors used by IoT agriculture applications. Then, the taxonomy is used to build an IoT agriculture dataset ontology based on OWL2, and a set of SWRL rules to reason about IoT dataset ontology individuals. Finally, the knowledge modeled in the proposed ontology is used to classify datasets by IoT agriculture applications using several ML classification techniques.

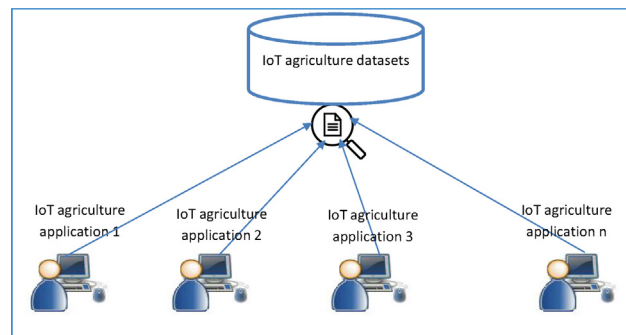


Fig. 1. Motivating scenario.

The remainder of the paper is organized as follows. Section 2 presents the related work. Section 3, describes the proposed IoT agriculture taxonomy. Section 4 discusses the IoT agriculture ontology development steps. Section 5 describes SWRL rules. Section 6 presents the classification process. Section 7 deals with the experiments. Results and discussions are presented in Sections 8. Finally, Section 9 concludes the paper and discusses future work.

2. Related work

The contributions proposed in this paper deal with several issues relating to taxonomies of application domains and dataset representation, agriculture ontologies, and ontology-based methods for classifying datasets.

2.1. Taxonomies and datasets representation

In this issue, a literature review based on query 1: “taxonomy” AND “iot”, query 2: “structure of dataset” and query 3: “dataset representation” has been conducted using PRISMA¹ guidelines (Fig. 2). It revealed several relevant papers:

The authors in (Rozsa et al., 2016) presented a taxonomy of application domains and sensor measurement types to identify and categorize the main sensors used in different IoT application domains. The analysis of several IoT-based contributions produced a taxonomy with three domains: Industrial, Smart Cities, and Healthcare.

A similar work (Pflanzner and Kertesz, 2018) proposes a taxonomy of IoT applications and IoT cloud applications composed of 12 categories and 64 properties in which five top-level categories have been revealed (context, participant, sensors, network and other). The context category constitutes the most important viewpoint in a great number of IoT applications. It determines the key features of an IoT application and its area of utilization. It is divided into five subcategories, namely: Body, building, city, industry, and environment.

A recent study by Rosero et al. (Rosero-Montalvo et al., 2022) introduced a taxonomy of IoT sensors. The authors classified sensors into three groups based on the sampling rate and how they transmit information to the microcontroller: analog sensors, digital sensors, and sensors based on communication protocol. The focus of the method was primarily on on-device data pre-processing and the use of filters to ensure the reliability of data uploaded to the cloud. However, the method may not address other crucial aspects such as IoT agriculture applications and dataset classification.

¹ <http://www.prisma-statement.org/PRISMAStatement/?AspxAutoDetectCookieSupport=1>.

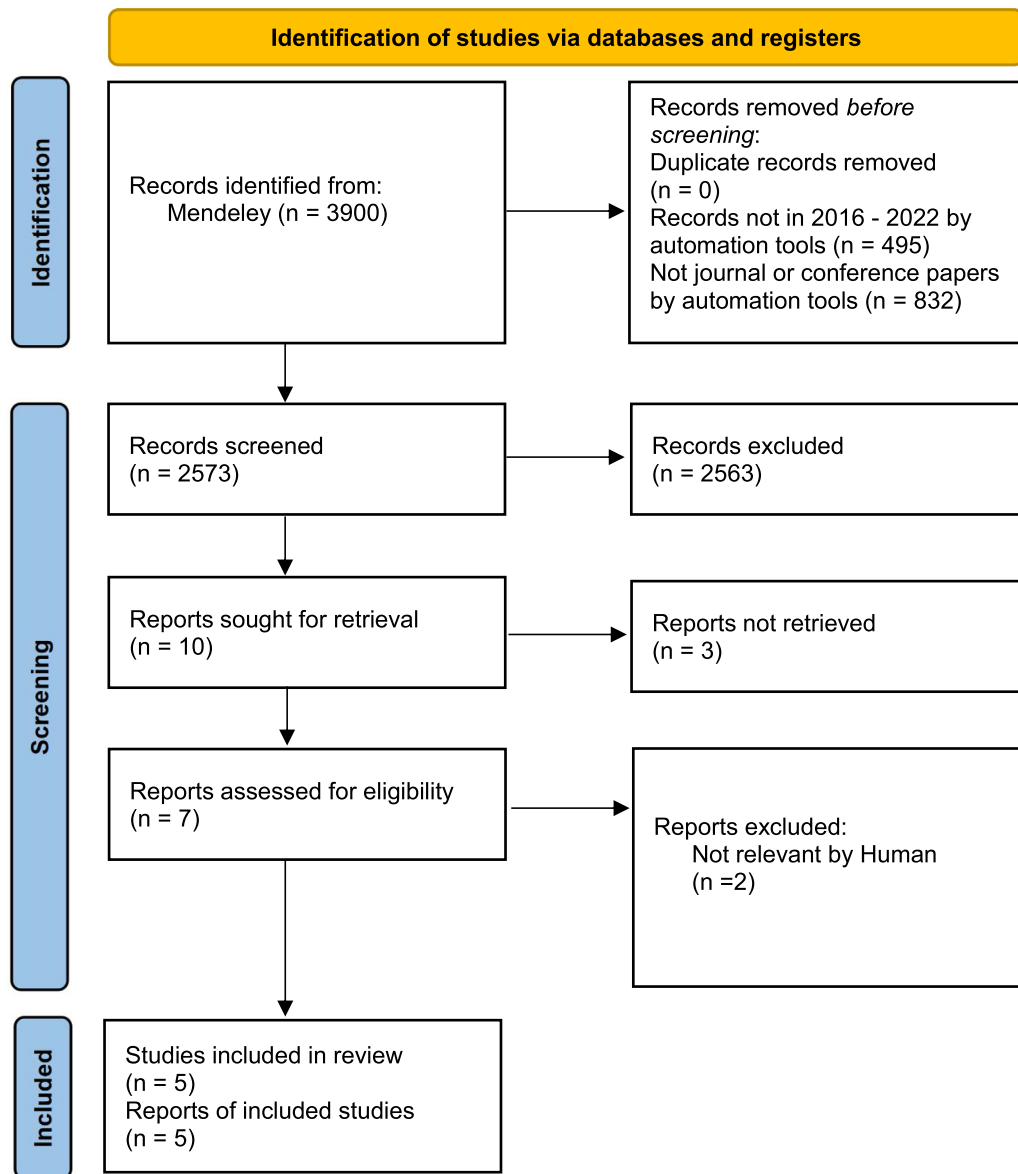


Fig. 2. Literature review on existing taxonomies and datasets representation based on PRISMA.

Rosli et al., from the University of Auckland, proposed a meta-model for describing the structure of datasets and the relationship between each concept. The meta-model was applied to existing public software engineering datasets (Rosli et al., 2016). The authors construct a precise definition for a measurement dataset and its potential elements. The proposed meta-model has included a set of elements such as value, label, metadata, structural, etc. It was evaluated with the existing measurement datasets from software repositories.

The authors of the paper (Wang et al., 2022) proposed a standardized representation of a dataset's metadata, called a data card, to establish a universal dataset metadata template. Each data card contains information such as the data owner, source location (or URL, for web data), date/time of collection, and file name. Metadata are stored in a key-value pair format to flexibly support the vast differences in metadata fields between data cards and addition of new data fields in the future. Their proposition allows to better capture hierarchical metadata such as table names, column names, etc. It is not

specific in agriculture domain and doesn't take into consideration the IoT.

Since the proposed taxonomies in the literature suffer from a lack of information on agriculture sensors applications. We propose, in this paper a new taxonomy of IoT agriculture sensors applications where we focus on the combination of sensors used in each application. The generated taxonomy is based on several reliable information sources, including research papers and published IoT agriculture datasets.

In other part, the concept of domain is crucial in classifying different datasets accurately and efficiently. The absence of this concept in the semantic model may hinder the ability to categorize datasets based on their specific domains, such as plant disease, irrigation, fertilization, or greenhouse cultivation.

In our work, we would provide a valuable means of organizing and categorizing datasets according to their relevant subdomains within agriculture. Our agriculture datasets representation is based on an analysis of the existing IoT agriculture applications used sensors in their implementations.

2.2. Agriculture ontology

As is to be expected, we can't propose an ontology for agriculture without taking a look at what already exists. In recent years, much work has focused on the use of ontologies in agriculture (Appendix A).

In (Jiang et al., 2020), authors proposed a new method to integrate heterogeneous data into knowledge bases that can support farmers in their activities, and to present global, real-time and comprehensive information to researchers. They gave a brief review on the semantic web technology applications for agriculture. However, their work is only at the design level.

The paper (Pani and Mishra, 2016) presented the building of an ontology in agriculture system in web ontology language (OWL). Their ontology is built with the data collected from India's various agriculture departments. Despite the fact that their work covers multiple areas of agriculture, they have omitted the IoT part.

A new ontology on agriculture extending AgriOnt is presented in the work (Fahad et al., 2021). The extended ontology ONTAGriX add new features like the scalability, the service-oriented and context awareness. Additionally, it incorporated the IoT subdomain and reasoning process for precision farming applications. It covers the domain of crops, fertilization, irrigation, pesticides, pests, soils. It is not designed to take dataset classification into consideration.

In (Deepa and Vigneshwari, 2022), authors proposed a new agricultural ontology focused on subdomains. It is designed to assist farmers using Jaccard relative extractor (JRE) and Naïve Bayes algorithm. The JRE is used to find the similarity between two sentences and words in the agricultural documents and the relations between two terms is identified via the Naïve Bayes algorithm. The sub-domains covered are weather, pest and soil. It doesn't take into consideration IoT and all sub-domains of agriculture. Despite combining ontology with machine learning, it's not a task of classifying datasets by domain.

A scheme for designing an ontology for agriculture domain is described in (Kaushik and Chatterjee, 2018). Their ontology is constructed by using automatic extraction of vocabulary and relationships between the terms. The lack of their work is that their algorithm does not use any dictionary or thesaurus specific to agricultural domains. So, the number of terms extracted is not large. They focused more on the crops sub-domain.

The work (Malik et al., 2021) presented the design and the development of a generic ontology for fertilizers. Despite the fact that their work is focused on a single domain and that IoT is absent from it, they validated their ontology with the help of a tool and a domain expert.

The contribution in (Deepa and Vigneshwari, 2019) is a novel Hybrid Neural for Agricultural Ontology Construction (HNN-AGOC). Their algorithm enables the automatic ontology construction for predicting crop over different climate. They use CNN in the algorithms for the classification of the data and RNN to predict the crop.

In (Santhia et al., 2016), authors developed a plants ontology framework. The ontology describes plants and his ingredients, helps to identify pests in crop and different types of diseases. Authors in (Symeonaki et al. 2022) proposed an ontology based IoT middleware to create a unique framework for livestock management. Their middleware encompasses the context-awareness approach. They tested their system in a real condition.

A new method of semantic linking and storing open government datasets of New Zealand's agriculture, land and rainfall based on the use of ontology is described in the paper (Kaur et al., 2022). The authors presented a method to link ontology manually (domain experts) and automatically.

The paper (Elumalai and Margret Anuncia, 2021) presented an ontology based knowledge for storing the details of the soil composition with different minerals. The ontology provides a structured formalized knowledge which is mined for different patterns that subsequently concludes with recommendation of crops and the soil composition suitable for crops.

Despite the availability of numerous agricultural ontologies, as far as our knowledge extends, there is currently no ontology specifically designed to facilitate the classification of datasets.

2.3. Ontology-based methods for classifying datasets

In this issue the conducted review, following the PRISMA guidelines, revealed some relevant papers (Appendix B).

In (Wang et al., 2020), authors investigated for classifying scientific datasets into research domains. They provided a new ontology selection task known as 'ontology classification' for finding the ontology which best describes a given dataset. One notable limitation of the method is the absence of the agricultural domain in the analysis. The paper covered the biomedical domain, the computer science domain, the physics domain, the finance domain, and the environment domain. Given the significant impact of agriculture on global economies and the increasing use of data-driven technologies in the agricultural sector, the exclusion of this domain restricts the applicability and practical relevance of the proposed ontology-based classification approach. The paper considered only five ontology candidates for classification: FIBO, UMLS, CSO, ENVO, and OPB + Physics. They only used the title and description of datasets for their classification task. The IoT is not taken into consideration in the work.

The second paper (Abu-Salih et al., 2018), is about a methodology for domain classification of users using semantic web and machine learning. The domains are political and non-political. The dataset contains data from twitter. This is a work on user classification, not dataset classification. The ontology is only used for feature extraction. While these ontologies cover a range of domains, the selection of ontologies might not represent the full diversity of research fields. Including a new ontology which represents datasets knowledge would provide a more comprehensive evaluation. Following the same perspectives, a new IoT agriculture ontology was developed to represent IoT agriculture datasets. The proposed ontology is characterized by considering dataset attributes, combination of sensors, and keywords of IoT agriculture sensors applications for the classification task.

3. IoT agriculture taxonomy

The first target of this paper is to propose a taxonomy that can be used as a foundation to classify IoT agricultural datasets. Indeed, a such taxonomy allows detecting the effect of combining various IoT agriculture sensors on identifying an IoT agriculture sensor application. This taxonomy aims to answer the following questions:

- Q1: Which IoT sensors are integrated in agriculture?
- Q2: What are the different IoT agricultural applications or sub-domains?
- Q3: Which sensors combinations are used in an IoT agricultural application?
- Q4: Can a single sensor generate more than one data type?

By answering the first question, we will be able to investigate the different IoT agriculture sensors in the field of smart agriculture. Question 2 allows identifying the existing IoT agricultural applications. Question 3, gives an insight into how the combination

of various IoT agriculture sensors can determine an IoT agricultural application. Finally, question 4 permits determining the different IoT agriculture sensors that can generate several data types.

3.1. Resources and frameworks review

To give responses to the above questions, a conducted review, following the PRISMA guidelines and based on query 1: “smart farming” and “iot”, query 2: “precision agriculture” and “iot”, query 3: “smart agriculture” and “iot”, query 4: “irrigation scheduling”, query 5: “water balance” and “iot”, query 6: “greenhouse management”, query 7: “plant disease” and “iot”, query 8: “fertigation” and “iot”, query 9: “soil health” and “iot”, query 10: “soil moisture” and “iot”; revealed seventy relevant papers from 2016 to 2022 (Fig. 3).

The selected research papers can be classified according to the IoT agricultural applications into four categories, namely: fertigation, greenhouse, irrigation, and plant disease presented respectively in Table 1.

The previous review shows that the majority of the studied papers relied heavily on weather-related sensors, including (rainfall, air temperature, air humidity, solar radiation, wind direction, wind speed, etc). Indeed, many agricultural operations are inextricably linked to the weather since it has a direct and immediate impact on irrigation, fertilization, planting, and growth operations. For that, the weather has been considered in this paper as a major part of IoT agriculture sensor applications. It is also worth mentioning that the combination of the different agricultural applications, conducts us to define another important agricultural application named “crop management”.

3.2. An IoT agriculture taxonomy

Based on the previous reviews, we propose a new IoT agriculture taxonomy that allows agricultural applications to choose the right dataset. We noticed that there are two types of sensors for agricultural applications, namely physical state sensors and chemical state sensors. The elements of the first type were collected from research papers while the others were gathered from published IoT agriculture datasets. The proposed taxonomy is shown in Tables 2 and 3.

Tables (Table 2 and Table 3) provide clear answers to the four questions designed in this paper: (i) There are several IoT sensors which can be integrated in agriculture and which have initially been identified in this study like: electrical conductivity, GPS, soil temperature, soil moisture, pH, turbidity, wind speed, pressure, air humidity, air temperature, etc (Fig. 4 and Fig. 5). (ii) This work has collected the six following sub-domains or applications in this field: crop management, irrigation management, fertigation management, greenhouse, plant disease and weather. (iii) We managed to reorganize the IoT sensors under the six agricultural sub-domains as follows:

- The crop management systems use the majority of IoT agriculture sensors.
- The irrigation management systems include a large number of sensors such as: soil temperature, soil moisture, ph, pressure, light, electrical conductivity, etc.
- The fertigation management systems incorporate the following sensor combination: nitrogen, phosphorous, magnesium, manganese, iron, aluminium, cobalt, nickel, potassium, calcium, bore, turbidity, sodium, ammonium, zinc, molybdenum, copper and carbon.
- The plant diseases sub-domain uses mainly the IoT agriculture sensors which incorporate location sensors such as: GPS sensors, satellite imagery and UAV, optical sensors like Hyperspec-

tral camera sensors, Laser sensors, Motion sensors, Multispectral camera sensors, Normal camera sensors, Thermal camera sensors and photoelectric sensors such as color sensors, light sensors and solar radiation sensors.

- Greenhouse management systems use basically gas sensors such as: propane, nitrogen dioxide, carbon monoxide, smoke and co2.
- The weather sub-domain uses: rainfall sensors, air temperature sensors, air humidity sensors, solar radiation sensors, wind direction sensors, wind speed sensors, wind gust sensors, dew-point sensors, etc.

(iv) Table 2 shows that the most commonly used IoT agriculture sensor is GPS, which is a location sensor that can give the following three data types: altitude, longitude and latitude; a finding that answers our research's fourth question.

The results shown in Tables 2 and 3 also involve that the different sub-domains are interrelated: the weather is integrated in all the other sub-domains, and the irrigation, fertigation and plant disease sub-domains are parts of the crop management system which is included in the greenhouse sub-domain.

4. An IoT agriculture dataset ontology

In order to design a domain ontology, it is necessary to follow a particular methodology. Several methodologies have been proposed in literature to develop the domain ontology, and guarantee the quality of the results of the design process (Sattar et al., 2020). The main purpose of the IoT agriculture dataset ontology is to share IoT agricultural datasets information between the users of IoT agricultural applications such as farmers or other interested persons. It helps in communication for better decision making, facilitating and improving the storage of information, and supporting the reuse of datasets. In our context, we focused on the most common steps in the majority of ontology development methodologies: (i) collection and analysis of the IoT agriculture datasets knowledge and (ii) Specification of the terminologies of the IoT Agriculture dataset ontology.

4.1. Collection and analysis of IoT agriculture datasets knowledge

Recently, numerous IoT agriculture datasets have been made publicly available (eg. Kaggle.com). These datasets vary in their attributes, label of the attribute, attribute's arrangements, temporal frequency, unit of measurement, etc. In the literature, a dataset can be defined as a collection of related data that can be manipulated as a unit by a machine. It appears to be unproblematic in its common use (Renear et al., 2010). The W3C Data Catalog Vocabulary DCAT² defines a dataset as “a collection of data, published or curated by a single agent”. Agriculture datasets come in many forms including numbers (for measuring ph, temperature or pressure), words (for measuring weather condition ‘normal, sunny, windy, rainy’), pixels or images (image of plant leaf disease), sound (sound of cattles) and other multi-media, and potentially other types, any of which might be collected by different IoT agriculture sensors. Each created dataset, has a dataset domain. That domain is defined as a collection of related observations with a specific and common area. The term dataset itself was not provided with a precise formal definition in the literature (Renear et al., 2010). This paper discusses the representation of IoT agriculture datasets knowledge through a meaningful structure. To this end, datasets and their elements must be precisely defined.

² <https://www.w3.org/TR/vocab-dcat/> examples-dataset-provenance (W3C Recommendation 04 February 2020-version 2).

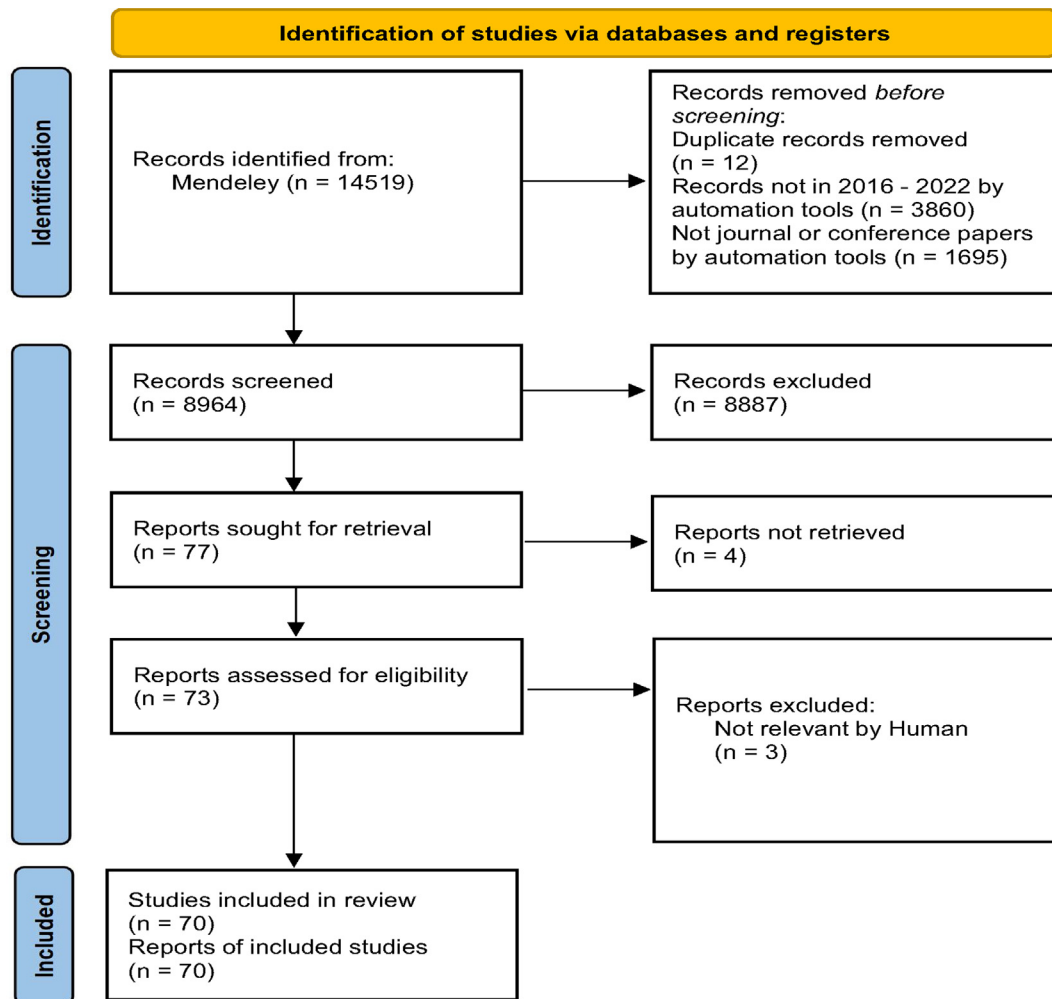


Fig. 3. Literature review on existing IoT agriculture systems.

Definition 1 (IoT Agriculture Sensor Dataset). An IoT agriculture sensor dataset can be defined as a collection of data describing the agriculture domain generated or not generated from different IoT agriculture sensors.

Example 1.

The agriculture & Weather – Kaggle (accessed on 25 October 2021) provides an agriculture and weather dataset. It contains the set of attributes: weather_temperature, weather_humidity, solar_radiation, soil_moisture, and soil temperature. Another source, Soil detection for cotton crop – Kaggle (accessed on 25 October 2021) provides a dataset for cotton crop soil detection. It contains attributes generated from IoT agriculture sensors: ph, temperature, humidity, density, electrical conductivity, N, P, K, Ca, Mg, and other attributes: grain surface, particle attached, particle width and cotton crop.

Definition 1 allows to represent an IoT agriculture sensor dataset through columns which contain sub-set columns generated by IoT agriculture sensors and named IoT agriculture dataset attributes.

Definition 2 (IoT Agriculture Sensors Subdomain or Application). An IoT agriculture sensors sub-domain represents an application of IoT sensors dataset that treats agriculture. IoT agriculture domain encompasses several IoT applications such as: crop management, irrigation scheduling, greenhouse management, fertigation management, plant disease forecasting and weather.

An IoT agriculture sensors application can be represented by a combination of labeled sensors (Tables 2, 3). It can also be defined

by keywords in which each IoT application has a different set of keywords (eg. Crop management = crop, motion, color).

Definition 3 (IoT agriculture Sensor). An IoT agriculture sensor is a smart, simple and small device that can sense or measure real conditions in the agricultural domain. In our work, an IoT agriculture sensor can be used in different IoT agricultural applications. An IoT agricultural application is defined as a combination of IoT agriculture sensors. Agriculture sensors are identified by names (temperature, humidity, ...) and can be classified into different types: Optical sensors, Location sensors, Electrochemical sensors, Mechanical sensors, Dielectric sensors, etc, (Table 4).

Definition 4 (IoT Agriculture Sensors Dataset Attribute). An IoT agriculture sensor dataset attribute can be defined as a key feature, output or a characteristic of a result produced by the IoT agriculture sensor. Attributes are considered as the key features that describe the IoT agriculture sensor dataset and they are related to a set of IoT agriculture sensors of a given IoT agriculture application. Every IoT agriculture sensor dataset attribute has a data type such as: soil moisture, temperature, soil color, environment humidity, leaf-wetness, electric conductivity, wind speed and direction, barometric pressure, carbon dioxide, ph value, light intensity, solar radiation, rainfall, size of crops, shape of crops, thickness of plant stem, altitude, longitude, latitude of the plants and others.

It has also a unit of measurement including lux for luminosity, m/s² for acceleration, Mg/L for humidity, m³ for water content, Fahrenheit (°F), the Celsius (°C), and the Kelvin (K) for temperature,

Table 1
IoT agriculture sensors used in IoT systems.

Papers	Sensor Type																								
	Air flow		Dielectric		Electrochemical				Location			Mechanical				Photoelectric			Optical				Temperature		
	pressure	Leaf wetness	Soil moisture	Electrical Conductivity	CO2	pH	Nitrogen	NPK	GPS	Satellite	UAV	Rainfall	Water level	Balance	Wind speed/direction	Color	Lihgt	Solar radiation	Camera	Thermal camera	Turbidity	Motion	Air humidity	Air Temperature	Soil temperature
Category 1: Fertigation Management Systems																									
(Mohanraj et al., 2018)			X														X					X	X		
(Hartanto et al., 2021)				X		X																			
(Placidi et al., 2021)			X																				X		
(Shamsi et al., 2020)				X		X																			
(Zubair and Adebiyi, 2022)			X				X																		
(Mohamed and Ahmed, 2018)			X	X		X																			
(Manickam, 2020)			X			X		X														X	X		
(Kulmány et al., 2022)			X																						
(Zhang et al., 2017)			X																			X	X	X	
(Sengupta et al., 2021)			X			X														X			X		
(Visconti et al., 2020)			X																			X	X		
(Jenal et al., 2021)			X									X													
(Raut et al., 2018)			X													X							X		
(Kiruthika et al., 2021)																		X							
(Zulhilmi et al., 2021)			X																						
(Haziq et al., 2022)			X			X																	X		
(Kumar et al., 2019)			X			X		X	X							X							X		
Category 2: Greenhouse Management Systems																									
(Abbassy and Ead, 2020)					X																	X	X		
(Hamouda and Elhabil, 2017)																						X	X		
(Proietti et al., 2021)	X															X		X				X	X		
(Jung et al., 2020)					X																	X	X		
(Ben Akka et al., 2019)	X		X		X																		X		
(Luo et al., 2016)																						X	X	X	
(Goel and Gurve, 2018)			X														X						X		
(Halim et al., 2017)					X												X					X	X		
(Rokade et al., 2022)			X		X												X					X	X		
(Rustia et al., 2020)																		X							
(Patil et al., 2022)			X									X					X					X	X		
(Li et al., 2017)																	X					X	X		
(Kiruthika et al., 2021)			X																			X	X		
(Chen et al., 2019)																						X	X		

(continued on next page)

Table 1 (continued)

Table 1 (continued)

Papers	Sensor Type													
	Air flow			Dielectric			Electrical			Electrochemical			Location	
	pressure	Leaf wetness	moisture	Soil	Electrical Conductivity	CO ₂	pH	Nitrogen	NPK	GPS	Satellite	UAV	Rainfall	Water level
Mechanical														
Balance														
Wind speed/direction														
Color														
Lihgt														
Solar radiation														
Photoelectric														
Camera														
Thermal camera														
Optical														
Turbidity														
Motion														
Air humidity														
Temperature														
Air Temperature														
Soil Temperature														
(Khatab et al., 2019)	X	X											X	
(Pourazar et al., 2019)												X		
(Nawaz et al., 2020)														
(Praneetha and Tabassum, 2021)														
(Pathan and Chakole, 2019)														
(Kumar et al., 2021)														

and other like 'mg_kg', 'mg', 'kg', 'percentage', 'perc', 'ppm', 'pascal', 'degree', 'km/h', 'km/s', 'mm', 'lux', 'ntu', 'm/s', 'meter', 'millimeter', '°', 'deg', 'f', 'mph', 'percent', etc.

Example 2.

In <https://www.kaggle.com/zohasohail/soil-detection-for-cotton-crop> temperature (degree Celsius), humidity (percentage), density (g/cm³), electric conductivity (ds/m), N(ratio), P(ratio), K (ratio), and Ca(ratio).

In the context of IoT agriculture a lot of abbreviations have been used to indicate attributes label. Some abbreviations have a weak semantic, and guide to bad comprehension by users when re-using datasets. To overcome this problem, we give for each attribute synonyms to improve the semantic of the attributes.

Example 3.

For the attribute mo=['mo', 'molybdenum', '%mo', 'perc_mo', 'mo_mg_kg', 'mo (ppm)'],

Example 4.

For the attribute c=['c', 'carbon', '%c', 'perc_c', 'c_mg_kg', 'c (ppm)'],

Example 5.

For the attribute radiation=['radiameter', 'radiation', 'sunshine'].

4.2. Specification of the terminologies of the IoT agriculture dataset ontology

As a result of the aforementioned definitions and in order to make the ontology formalization step easier, we give an IoT agriculture dataset meta-model (Fig. 6) based on UML class diagram to represent the different elements of the dataset and their relationships. The concepts defined in the previous activity are specified according to the class diagram elements. Likewise, we used the description presented by W3C DCAT to complete sensor dataset properties such as the identifier, the name of the dataset, its path, the description, and its version.

The meta-model is used as a sufficient way to unify and guide the formalization of IoT agriculture dataset ontology. It represents the basis step to construct OWL ontology. Meta-model concepts are specified in the OWL's components. An OWL ontology is defined as a set of classes, object properties linking classes together, restrictions on properties and axioms (Maedche and Staab, 2001). Based on both OWL ontology definition and the given class diagram, we can construct the dataset ontology, in which the classes of the meta-model constitute the main OWL Classes and sub-classes (Tables 5, 6). The Relationships between IoT agriculture dataset elements are represented through a set of object properties by specifying the domain and the range of each one (Table 7). To represent the attributes of different classes, we used data properties. Ontologies used data properties to connect individuals with literals.

The most important elements of the proposed ontology will be: dataset, domain, sensor, attribute, column, datatype and unit (Table 5).

The most important subclasses will be: AirFlowSensor, DielectricSensor, ElectroChemicalSensor, LocationSensor, MechanicalSensor, OpticalSensor, PhotoElectricSensor and TemperatureSensor (Table 6).

The most important relationships will be: hasAttribute, hasDatatype, hasDomain, hasOutput, hasRelative, hasSynonym, hasUnity, hasUse and isComposed (Table 7).

5. SWRL rules

The inference is the ability to make deductions about instances, regarding the classes, properties and restrictions defined (Krima et al., 2009). Using semantic technologies, we can enrich our semantic structure and reason on it to infer and create new knowl-

Table 2

IoT agriculture taxonomy based on research papers.

IoT sensor	Crop	Fertigation	Greenhouse	Irrigation	Plant disease	Weather
Electrical Conductivity	X	X	X	X	X	
GPS	X	X	X	X	X	X
Soil Temperature	X	X	X	X		
Soil Moisture	X	X	X	X	X	
pH	X	X	X	X	X	
Turbidity	X	X			X	
Wind Speed	X			X	X	X
Pressure			X	X		
Humidity	X		X	X	X	X
Temperature	X		X	X	X	X
Light			X	X		X
Rainfall	X		X	X	X	X
Carbon dioxide			X			
Motion			X		X	
Solar radiation	X		X	X	X	X
Satellite imagery	X		X	X	X	X
Wind direction	X		X	X	X	X
Color	X		X		X	
Leaf wetness	X				X	
Normal camera	X		X	X	X	
Thermal camera	X			X		X
Multispectral camera	X				X	X
Hyperspectral camera	X				X	X
Water level	X	X	X	X	X	
Nitrogen		X				
Ultrasonic				X		
Balance				X		

Table 3

An expanded IoT agriculture taxonomy based on published IoT agriculture datasets.

IoT sensor	Crop	Fertigation	Greenhouse	Irrigation	Plant disease	Weather
Magnesium	X	X				
Iron	X	X				
Nickel	X	X				
Bore	X	X				
Zinc	X	X				
Colbalt	X	X				
Carbon	X	X				
Manganese	X	X				
Sodium	X	X				
Ammonium	X	X				
Cooper	X	X				
Molybdenum	X	X				
Calcium	X	X				
Aluminium	X	X				
Propane			X			
Carbon monoxide			X			
Nitrogen dioxide			X			
Heat index	X		X	X	X	X
Smoke			X			
Phosphorus		X				
Potassium		X				
Dewpoint	X		X	X	X	X
Wind Gust	X		X	X	X	X

edge. To this end, SWRL (Semantic Web Rule Language) is well adopted. SWRL is based on OWL-DL. It is the most appropriate rule language for ontology-based models. It provides association rules, and allows to associate instances to new classes and to create properties between instances (Fortineau et al., 2012). SWRL rules can make a semantic representation more complete (Djakhidjakhia et al., 2019). Based on the IoT dataset ontology elements, we give two important rules:

$R1 : Dataset(?x), isComposed(?x, ?a), isRelative(?a, ?b), Sensor(?b), hasOutput(?b, ?y) \rightarrow hasAttribute(?x, ?y)$

(1)

$R2 : Attribute(?x), isOutput(?x, ?a), hasRelative(?a, ?y)$

$\rightarrow hasSynonym(?x, ?y)$ (2)

The first rule $R1$ enables to filter the list of IoT agriculture dataset attributes related to IoT sensors. It indicates that if a dataset(x) is composed of column a and column a is relative to the sensor b and this sensor has as output an attribute y then the dataset x has y as an attribute.

The second rule $R2$ enables to identify synonyms. It indicates that if an attribute x is defined as an output of sensor a and this one has a relationship with column y then x and y are synonyms.

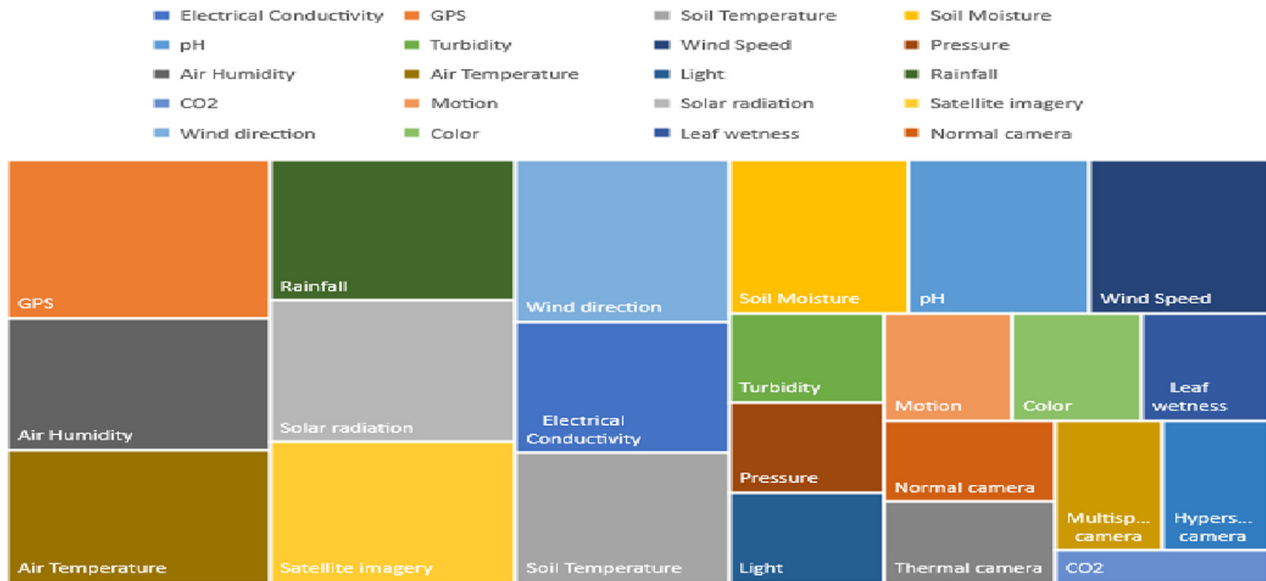


Fig. 4. Illustration of sensor used in IoT agricultural applications.



Fig. 5. Illustration of sensor used in IoT agricultural datasets.

Table 4

Classification of agriculture sensors according to their types.

Sensors Types	IoT Agriculture Sensors
Air flow sensor	Pressure sensor
Dielectric sensor	Leaf wetness sensor, Soil moisture sensor
Electrochemical sensor	Aluminium sensor, Ammonium sensor, Bore sensor, Calcium sensor, Carbon monoxide sensor, Carbon sensor, CO2 sensor, Cobalt sensor, Copper sensor, Electrical conductivity sensor, Iron sensor, Maganese sensor, Magnesium sensor, Molybdenum sensor, Nickel sensor, Nitrogen sensor, Phosphorous sensor, ph sensor, Potassium sensor, Propane sensor, Smoke sensor, Sodium sensor, Zink sensor
Location sensor	GPS sensor, Satellite Imagery, UAV
Mechanical sensor	Rainfall sensor, Wind direction sensor, Wind gust sensor, Wind speed sensor
Optical sensor	Hyperspectral camera sensor, Laser sensor, Motion sensor, Multispectral camera sensor, Normal camera sensor, Thermal camera sensor, turbidity sensor.
Photoelectric sensor	Color sensor, Lihgt sensor, Solar radiation sensor
Temperature sensor	Air humidity sensor, Air temperature sensor, dewpoint sensor, heat Index sensor, Soil temperature sensor

6. The process of the IoT agriculture datasets classifier

After the development of the IoT dataset ontology, we propose a process that allows to classify the IoT agriculture datasets (Fig. 7).

The classification process for a new dataset is as follows:

- Loading the new agricultural IoT dataset
- Extract dataset characteristics (creation date, owner, column names, ...)
- Retrieve from the dataset ontology the list of attributes for each sensor, along with its semantic and syntactic synonyms.
- For each dataset feature (column name), we identify which sensors attribute it corresponds to by calculating its similarity to the list of attributes retrieved from the ontology.
- The next step is to construct the feature vector. For each attribute of an identified sensor or keyword, in the feature vector it is assigned the value 1, otherwise 0. So if the dataset has no characteristic corresponding to an attribute or contains no keyword in our taxonomy, it will have as characteristic vector: [0,0,0, 0].
- Predict the dataset domain using the classification algorithm.

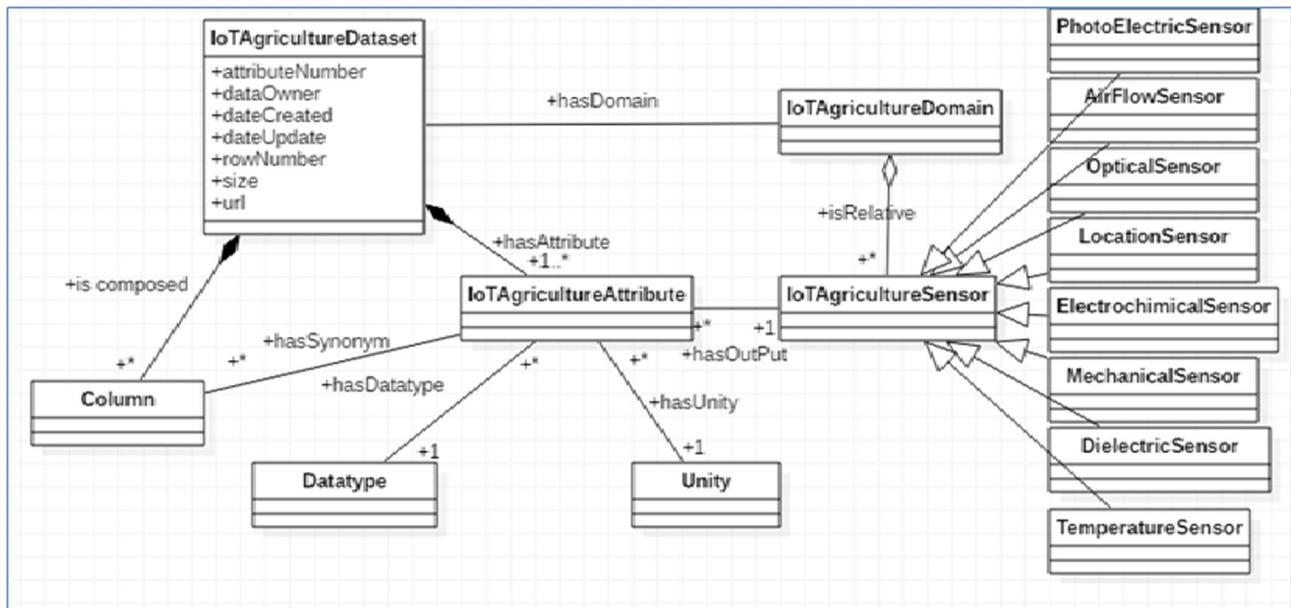


Fig. 6. UML class diagram of the IoT agriculture sensor dataset meta-model.

Table 5
Defined OWL classes.

OWL Class	Description
Agricultural Dataset	Is the main class it has data properties including: hasAttributeNumber, hasDataOwner; hasDateCreated, hasDateUpdaed, hasRowNumber, hasSize, hasUrl
Sensor Attribute	The set of dataset attributes related to an IoT sensor
Column	The set of all dataset attributes.
Sensor Datatype	it represents the datatype of an IoT sensor data
Agricultural Domain	It represents an IoT agricultural domain
Agricultural Sensor	It represents an IoT agriculture sensor
Unity	It represents the unity of measurement

- Instantiate the dataset in the ontology with all its characteristics and its domain too. The dataset is considered an instance of the Dataset class in the ontology. Its metadata, such as creation date, owner, size, etc., are considered attributes of the Dataset class. Column names are considered instances of the Column class. The isComposed relationship links the dataset to its column names. The dataset domain is specified by linking to an existing domain via the hasDomain relationship.
- New knowledge is inferred from the ontology. With the first SWRL rule, we associate the dataset with the set of sensors it use. The second allows us to expand our syntactic and semantic synonym dictionary. Each instantiation of a new dataset is an opportunity to add new vocabularies to our synonym base. In this way, our taxonomy is evolutionary.

7. Experiments

In this work, IoT agriculture dataset ontology and SWRL have been implemented automatically taking the advantage of Python. The implementation of dataset ontology produces an OWL file ([Appendix C](#)). The SWRL rules created are also stored in the same OWL file ([Appendix D](#)). The OWL file can be loaded in Protege ([Fig. 8](#)).

The relationships between IoT agriculture domains are specified by Python. The following figure depicts that the weather is included in the irrigation domain (Fig. 9):

We used our group of test datasets to evaluate the performance of our approach.

Then, the process extracts the features' vector and executes the classifier. The classification result is as well added to dataset ontology as a new property "hasDomain irrigation" (Fig. 10).

Fig. 10 shows that 'Dataset_sensors.Csv' is classified as dataset for the IoT irrigation systems. (see Fig. 11).

It is composed of: solar_radiation, soil_temperature, weather_humidity, soil_moisture, weather_temperature. Using SWRL rules, we infer `hasSynonym(weather_humidity, air_humidity)`, `hasSynonym(weather_temperature, air_temperature)` as well as, we infer the list of dataset attributes related to IoT agriculture sensors: `hasAttribute(Dataset_sensors, solar_radiation)`, `hasAttribute(Dataset_sensors, air_humidity)`, `hasAttribute(Dataset_sensors, soil_temperature)`, `hasAttribute(Dataset_sensors, soil_moisture)`, `hasAttribute(Dataset_sensors, weather_temperature)`.

Table 7
Defined relationship.

Object property	Domain	Range
hasAttribute	Dataset	Attribute
hasDatatype	Attribute	Datatype
hasDomain	Dataset	Domain
hasOutput	Sensor	Attribute
hasRelative	Sensor	Column
hasSynonym	Attribute	Column
hasUnity	Attribute	Unity
hasUse	Domain	Sensor
isComposed	Dataset	Column

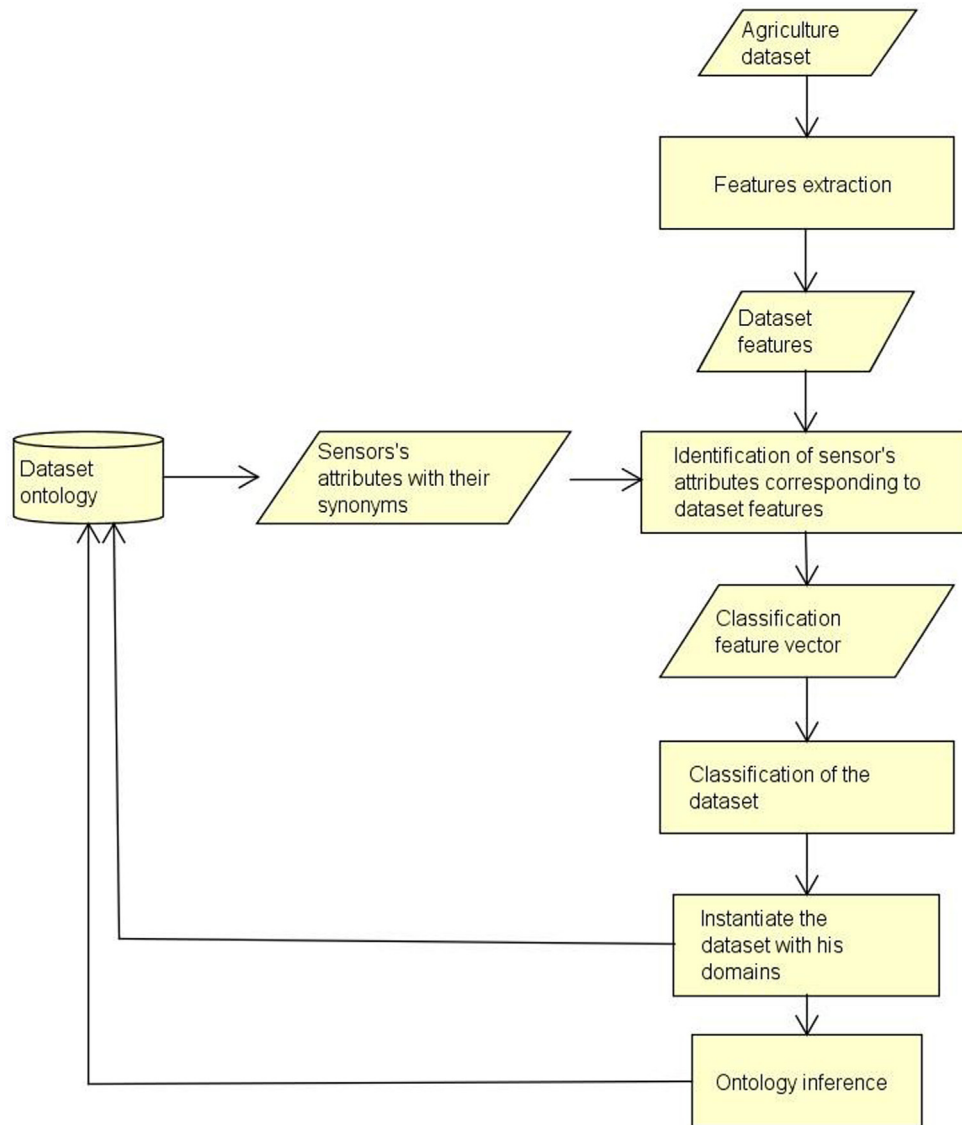


Fig. 7. The process of the IoT agriculture datasets classifier.

sors, air_temperature), hasAttribute (Dataset_sensors, soil_temperature) and hasAttribute (Dataset_sensors, soil_moisture). Identified attributes constitute the basis to construct the feature classification vector.

8. Results and discussion

We employed the popular sklearn library in Python for the classification task, and the sklearn.metrics module was utilized for comparing the performance of various machine learning algorithms. Table 8 presents a comparison of different metrics, including precision, recall, accuracy, and F1-score, obtained from the classification process within the IoT dataset ontology.

Based on the results (Table 8), it is evident that Linear Discriminant Analysis, Decision Tree, Gaussian Naive Bayes, Random Forest, and Support Vector Machine (SVM) exhibit good performance, with a relatively small gap between them. These algorithms demonstrate promising potential for the classification task in the context of our study.

Fig. 8 illustrates the accuracy values achieved by different machine learning algorithms, highlighting that SVM and Decision Tree outperformed the other algorithms in terms of accuracy.

To demonstrate the impact of the proposed approach on the quality of the results, we conducted an additional analysis. We repeated the evaluation using the same machine learning algorithms but excluded the proposed ontology. The results (Table 8) clearly indicate that the incorporation of the ontology significantly enhances the classifier's performance. On average, there was a notable gain of 0.34% in accuracy and 0.21% in the F-score when utilizing the ontology in comparison to the results obtained without it.

These findings highlight the importance and effectiveness of incorporating the proposed ontology in the classification process. The ontology aids in capturing domain-specific knowledge and enhancing the representation and understanding of agricultural concepts. By leveraging the ontology, the classifier demonstrates improved accuracy and F-score, indicating its ability to make more accurate predictions and achieve better overall performance.

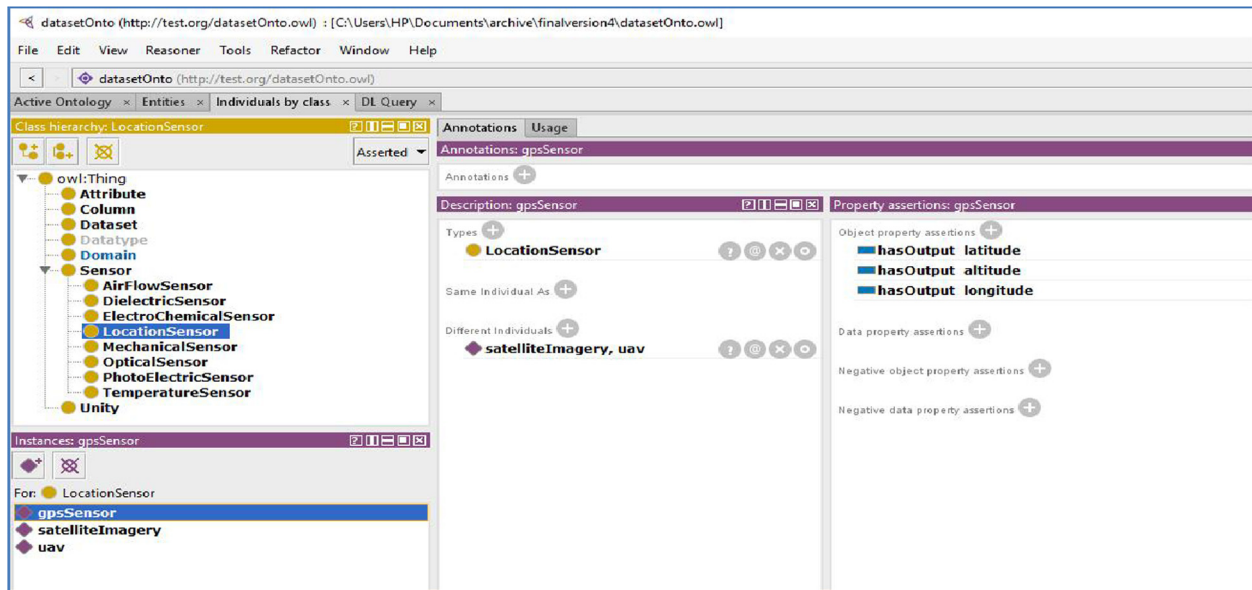


Fig. 8. Visualization of IoT dataset ontology in protege.

```
with datasetOnto:
    irrigation.hasUse=[DielectricSensor("soilMoistureSensor"),
        TemperatureSensor("soilTemperatureSensor"),
        AirFlowSensor("pressureSensor"),
        PhotoElectricSensor("lightSensor"),
        ElectroChemicalSensor("electricalConductivitySensor"),
        ElectroChemicalSensor("phSensor")]

    domain1=Domain("weather")
    for c in domain1.hasUse: irrigation.hasUse.append(c)
```

Fig. 9. Implementation of irrigation domain.

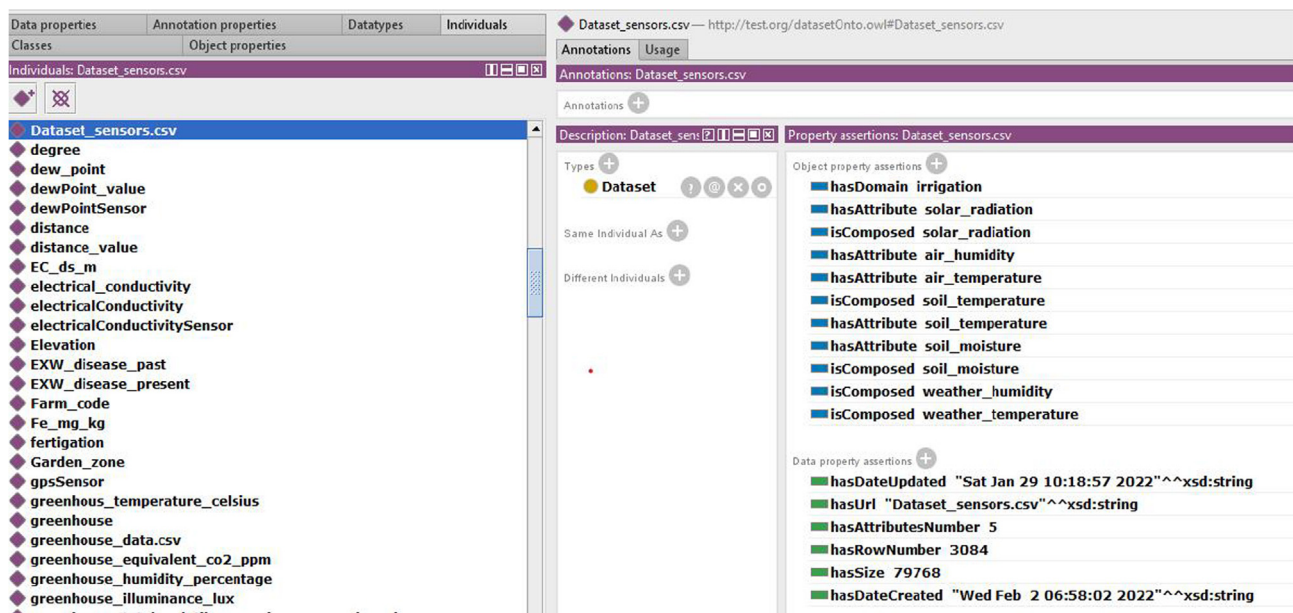


Fig. 10. Classification result of Dataset_sensors.csv.

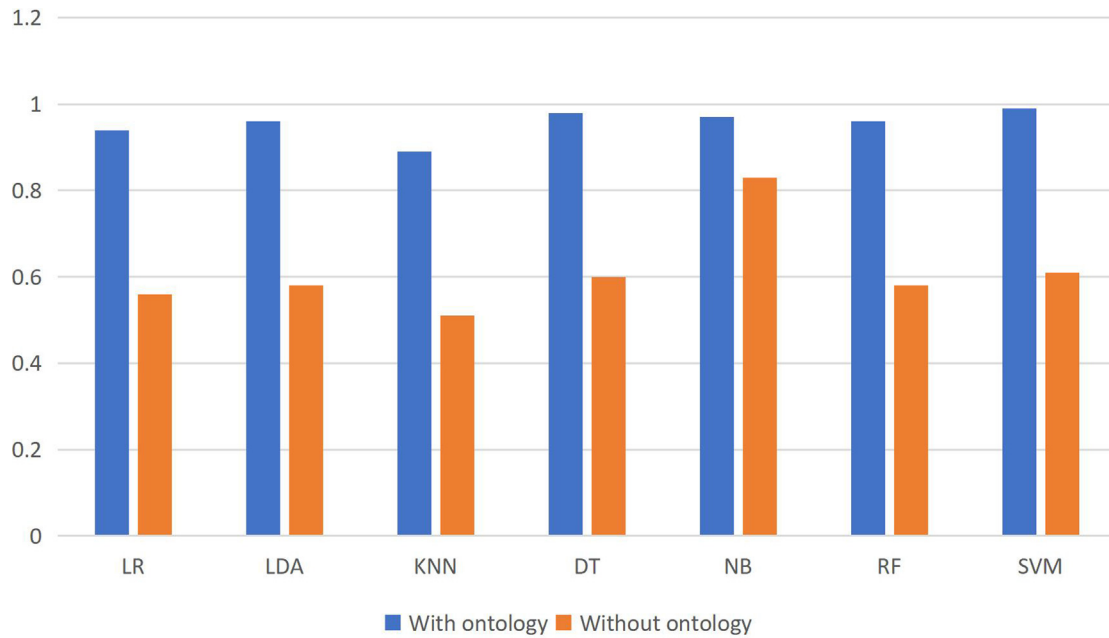


Fig. 11. Comparison of classification accuracy.

Table 8

Classification results with and without dataset ontology.

Algorithm	Precision with Onto	Precision without	Recall with	Recall without	Accuracy with	Accuracy without	F-score with	F-score without
Logistiques Regression (LR)	0.59	0.39	0.67	0.45	0.94	0.56	0.62	0.42
Linear Discriminant Analysis (LDA)	0.94	0.71	0.85	0.64	0.96	0.58	0.84	0.59
K-Nearest Neighbors (KNN)	0.73	0.52	0.59	0.37	0.89	0.51	0.63	0.40
Decision Tree (DT)	0.94	0.71	0.96	0.75	0.98	0.60	0.95	0.70
Gaussian Naive Bayes (NB)	0.94	0.88	0.88	0.80	0.97	0.83	0.89	0.81
Random Forest (RF)	0.95	0.72	0.88	0.67	0.96	0.58	0.88	0.63
Support Vector Machine (SVM)	0.92	0.69	0.96	0.75	0.99	0.61	0.93	0.68
Average	0.85	0.66	0.82	0.63	0.95	0.61	0.82	0.60

9. Conclusion

The main research contribution of this paper is the proposition of a semantic classification method of IoT agriculture datasets by the combination of the web semantic technologies with Machine Learning algorithms.

First of all, we have proposed a taxonomy of the IoT agriculture domain based on a literature review which identified several IoT agriculture domains, we have identified for each domain a combination of IoT agriculture sensors. The proposed taxonomy is used in the construction of the dataset ontology as well as the classification of IoT datasets.

Then, we have proposed an extensible domain ontology. This ontology describes IoT agriculture dataset elements based on the collection and the analysis of activities as well as a set of suggested definitions to complete and improve the quality of the proposed extensible structure. We mean by extensible that other sensor types and IoT agriculture applications can be identified and added. Based on the power and expressivity of OWL and SWRL, we have inferred new knowledge that improved the quality of results. Dataset ontology supports the re-use of IoT agriculture datasets. It improves the storage of all IoT datasets with a complete description of its properties (URL, size, row number, etc.), and its attributes related to IoT sensors. The description completes also information on dataset attributes like: unity of measurement and data type,

synonyms of attributes and domain keywords. It facilitates the user comprehension when re-using IoT datasets.

Finally, we have used machine learning algorithms for classifying IoT datasets. They appeared as a relevant method to enhance IoT agriculture datasets classification. The experiments show that the background knowledge provided by dataset ontology and the optimization of features have improved the prediction of the results and the performance of the classification. The results have shown that it is possible to classify IoT agriculture datasets with high performances and accuracy (0.98) and (0.99) using Decision tree classifier and SVM respectively.

However, the proposed method fails to classify image datasets which is considered as another challenge to complete this work. To overcome this problem, it is necessary to adopt other mechanisms such as image processing.

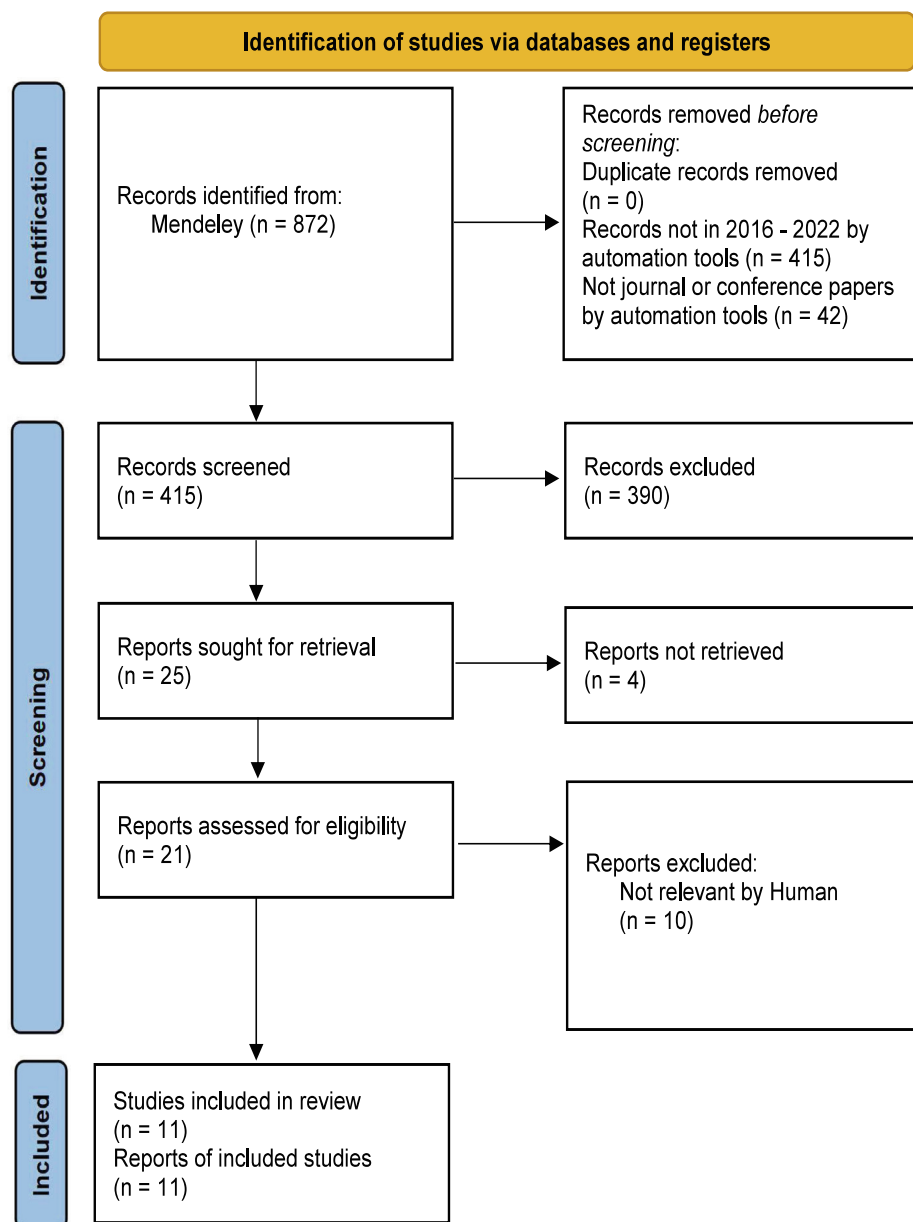
As a future work, we plan to analyze, represent and classify other types of datasets including datasets that contain images. Also, we plan to more extend and complete the proposed dataset ontology by analyzing the value interval of attributes

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

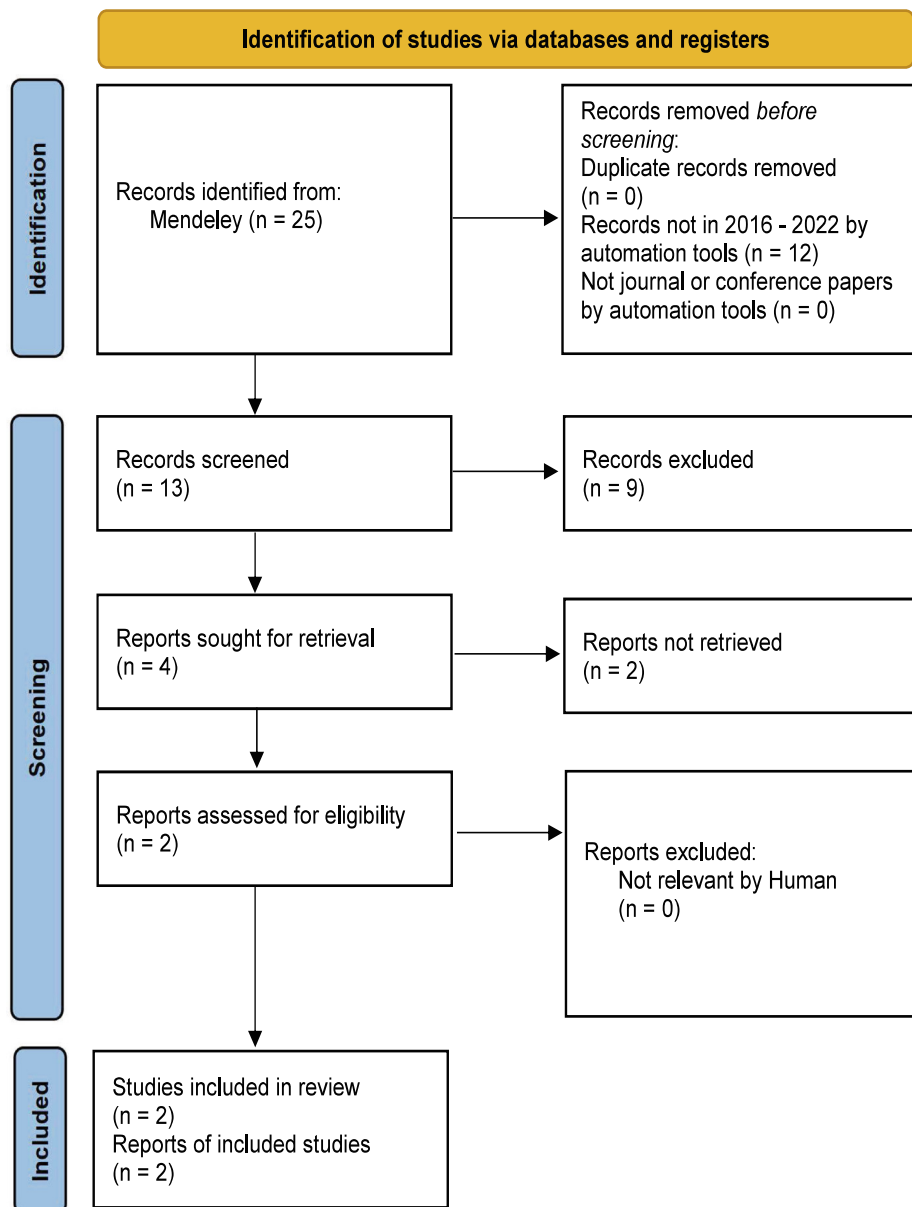
Appendix A

Literature Review on Agriculture Ontologies based on the query (“agriculture” and “ontology”) and flows PRISMA Guidelines.



Appendix B

Literature Review on Ontology-based Methods for Classifying Datasets based on the query (“domain classification” AND “ontology”) and follows PRISMA Guidelines.



Appendix C

Part OWL file of the Dataset ontology Created with Python.

```

1
2 <!-- http://test.org/datasetOnto.owl#AirFlowSensor -->
3 <owl:Class rdf:about="&datasetOnto;AirFlowSensor">
4   <rdfs:subClassOf rdf:resource="&datasetOnto;Sensor"/>
5 </owl:Class>
6
7 <!-- http://test.org/datasetOnto.owl#Attribute -->
8 <owl:Class rdf:about="&datasetOnto;Attribute">
9   <rdfs:subClassOf rdf:resource="&owl;Thing"/>
10 </owl:Class>
11
12 <!-- http://test.org/datasetOnto.owl#Column -->
13 <owl:Class rdf:about="&datasetOnto;Column">
14   <rdfs:subClassOf rdf:resource="&owl;Thing"/>
15 </owl:Class>
16
17 <!-- http://test.org/datasetOnto.owl#Dataset -->
18 <owl:Class rdf:about="&datasetOnto;Dataset">
19   <rdfs:subClassOf rdf:resource="&owl;Thing"/>
20 </owl:Class>
21
22 <!-- http://test.org/datasetOnto.owl#Datatype -->
23 <owl:Class rdf:about="&datasetOnto;Datatype">
24   <rdfs:subClassOf rdf:resource="&owl;Thing"/>
25 </owl:Class>
26
27 <!-- http://test.org/datasetOnto.owl#DielectricSensor -->
28 <owl:Class rdf:about="&datasetOnto;DielectricSensor">
29   <rdfs:subClassOf rdf:resource="&datasetOnto;Sensor"/>
30 </owl:Class>
31
32 <!-- http://test.org/datasetOnto.owl#Domain -->
33 <owl:Class rdf:about="&datasetOnto;Domain">
34   <rdfs:subClassOf rdf:resource="&owl;Thing"/>
35 </owl:Class>
36
37 <!-- http://test.org/datasetOnto.owl#ElectroChemicalSensor -->
38 <owl:Class rdf:about="&datasetOnto;ElectroChemicalSensor">
39   <rdfs:subClassOf rdf:resource="&datasetOnto;Sensor"/>
40 </owl:Class>
41
42 <!-- http://test.org/datasetOnto.owl#LocationSensor -->
43 <owl:Class rdf:about="&datasetOnto;LocationSensor">
44   <rdfs:subClassOf rdf:resource="&datasetOnto;Sensor"/>
45 </owl:Class>
46
47 <!-- http://test.org/datasetOnto.owl#MechanicalSensor -->
48 <owl:Class rdf:about="&datasetOnto;MechanicalSensor">
49   <rdfs:subClassOf rdf:resource="&datasetOnto;Sensor"/>
50 </owl:Class>
51
52 <!-- http://test.org/datasetOnto.owl#OpticalSensor -->
53 <owl:Class rdf:about="&datasetOnto;OpticalSensor">
54   <rdfs:subClassOf rdf:resource="&datasetOnto;Sensor"/>

```

```

55     </owl:Class>
56
57     <!-- http://test.org/datasetOnto.owl#PhotoElectricSensor -->
58     <owl:Class rdf:about="#datasetOnto;PhotoElectricSensor">
59         <rdfs:subClassOf rdf:resource="#datasetOnto;Sensor"/>
60     </owl:Class>
61
62     <!-- http://test.org/datasetOnto.owl#Sensor -->
63     <owl:Class rdf:about="#datasetOnto;Sensor">
64         <rdfs:subClassOf rdf:resource="#owl;Thing"/>
65     </owl:Class>
66
67     <!-- http://test.org/datasetOnto.owl#TemperatureSensor -->
68     <owl:Class rdf:about="#datasetOnto;TemperatureSensor">
69         <rdfs:subClassOf rdf:resource="#datasetOnto;Sensor"/>
70     </owl:Class>
71
72     <!-- http://test.org/datasetOnto.owl#Unity -->
73     <owl:Class rdf:about="#datasetOnto;Unity">
74         <rdfs:subClassOf rdf:resource="#owl;Thing"/>
75     </owl:Class>
76

```

Appendix D

Part of the OWL File of SWRL created by Python.

```

1  :91 <http://www.w3.org/2003/11/swrl#head> :91 .
2  :91 <http://www.w3.org/1999/02/22-rdf-syntax-ns#first> :85 .
3  :91 <http://www.w3.org/1999/02/22-rdf-syntax-ns#rest> <http://www.w3.org/1999/02/22-rdf-syntax-ns#nil> .
4  :92 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://www.w3.org/2003/11/swrl#Imp> .
5  :93 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://www.w3.org/2003/11/swrl#ClassAtom> .
6  :93 <http://www.w3.org/2003/11/swrl#classPredicate> <http://test.org/datasetOnto.owl#Attribute> .
7  :93 <http://www.w3.org/2003/11/swrl#argument1> <urn:swrl#x> .
8  :94 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> .
9  :94 <http://www.w3.org/2003/11/swrl#propertyPredicate> <http://test.org/datasetOnto.owl#isOutput> .
10 :94 <http://www.w3.org/2003/11/swrl#argument1> <urn:swrl#x> .
11 :94 <http://www.w3.org/2003/11/swrl#argument2> <urn:swrl#a> .
12 :95 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> .
13 :95 <http://www.w3.org/2003/11/swrl#propertyPredicate> <http://test.org/datasetOnto.owl#hasRelative> .
14 :95 <http://www.w3.org/2003/11/swrl#argument1> <urn:swrl#a> .
15 :95 <http://www.w3.org/2003/11/swrl#argument2> <urn:swrl#y> .
16 :96 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> .
17 :96 <http://www.w3.org/2003/11/swrl#propertyPredicate> <http://test.org/datasetOnto.owl#hasSynonym> .
18 :96 <http://www.w3.org/2003/11/swrl#argument1> <urn:swrl#x> .
19 :96 <http://www.w3.org/2003/11/swrl#argument2> <urn:swrl#y> .
20 :92 <http://www.w3.org/2003/11/swrl#body> :97 .
21 :97 <http://www.w3.org/1999/02/22-rdf-syntax-ns#first> :93 .
22 :97 <http://www.w3.org/1999/02/22-rdf-syntax-ns#rest> :98 .
23 :98 <http://www.w3.org/1999/02/22-rdf-syntax-ns#first> :94 .
24 :98 <http://www.w3.org/1999/02/22-rdf-syntax-ns#rest> :99 .
25 :99 <http://www.w3.org/1999/02/22-rdf-syntax-ns#first> :95 .
26 :99 <http://www.w3.org/1999/02/22-rdf-syntax-ns#rest> <http://www.w3.org/1999/02/22-rdf-syntax-ns#nil> .
27

```

Appendix E

Result of the Inference Step.

```
with datasetOnto:
  sync_reasoner_pellet(infer_property_values = True,infer_data_property_values = True)

* Owlready2 * Running Pellet...
java -Xmx2000M -cp C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\antlr-3.2.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\antlr-runtime-3.2.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\aterm-java-1.6.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\commons-codec-1.6.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\httpclient-4.2.3.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\httpcore-4.2.2.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\jcl-over-slf4j-1.6.4.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\jena-arq-2.10.0.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\jena-core-2.10.0.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\jena-iri-0.9.5.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\jena-rdb-0.10.0.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\jgrapht-jdk1.5.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\log4j-1.2.16.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\owlapi-distribution-3.4.3-bin.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\pellet-2.1.1.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\slf4j-api-1.6.4.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\slf4j-log4j12-1.6.4.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\vercesmp1-2.10.0.jar;C:\Users\WP\anaconda3\lib\site-packages\owlready2\pellet\xml-api-1.4.01.jar pellet.PelletReasoner --loader Jena --input-format N-Triples --infer-prop-values --infer-data-prop-values --ignore-imports C:\Users\WP\AppData\Local\Temp\tmp3bd7rg4d

* Owlready2 * Adding relation datasetOnto.solar_radiation hasSynonym datasetOnto.solar_radiation
* Owlready2 * Adding relation datasetOnto.Dataset_sensors.csv hasAttribute datasetOnto.soil_temperature
* Owlready2 * Adding relation datasetOnto.Dataset_sensors.csv hasAttribute datasetOnto.soil_moisture
* Owlready2 * Adding relation datasetOnto.Dataset_sensors.csv hasAttribute datasetOnto.solar_radiation
* Owlready2 * Adding relation datasetOnto.soil_temperature hasSynonym datasetOnto.soil_temperature
* Owlready2 * Adding relation datasetOnto.soil_moisture hasSynonym datasetOnto.soil_moisture
```

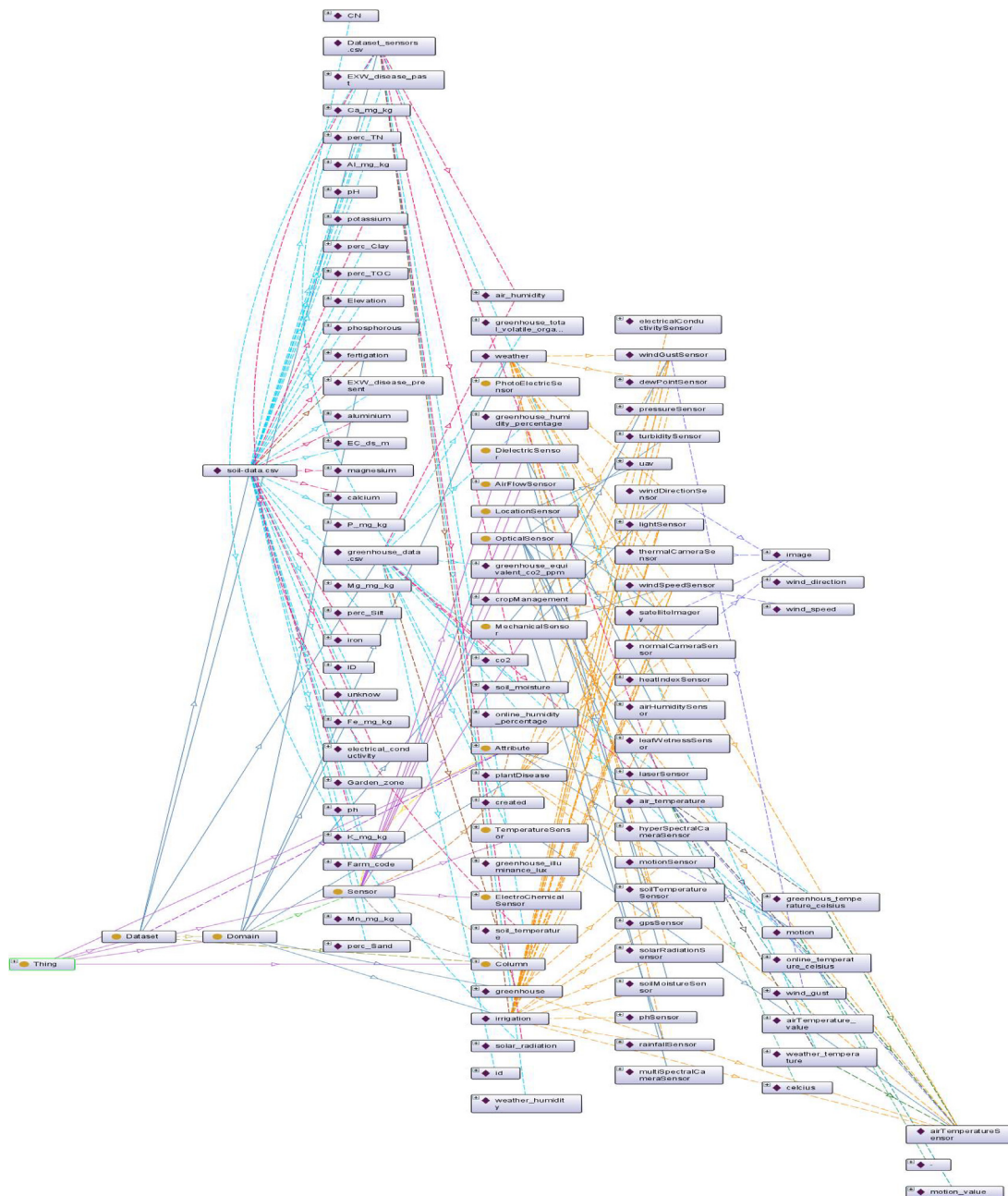
Appendix F

Summarizing of the relevant ontology metrics.

Ontology metrics	
Metrics	
Axiom	884
Logical axiom count	652
Declaration axioms count	232
Class count	16
Object property count	11
Data property count	7
Individual count	199

Appendix G

Result of the Inference Step.



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